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Chapter 1

General Introduction

1.1 General Context

Financial markets process trillions of dollars in transactions daily across equities, currencies, commodities, and digital assets. Over the past two decades, algorithmic trading has evolved from a niche practice used by quantitative hedge funds into the dominant force in global markets, now accounting for the majority of daily trading volume on major exchanges.

The core appeal is clear: algorithms process data at speeds no human can match and are immune to emotional bias. Yet building a consistently profitable system remains one of the hardest problems in applied machine learning. Markets are non-stationary — their statistical properties shift over time. A strategy that excels during a trending bull market may fail entirely in a volatile, mean-reverting environment. This sensitivity to changing market conditions is not a minor detail; it is the central challenge of the field.

Early rule-based systems lacked adaptability. Machine learning, and reinforcement learning in particular, offered a path forward by framing trading as a sequential decision problem. More recently, large language models introduced the ability to reason over unstructured data — news, macroeconomic commentary, sentiment — opening a new dimension in trading research. However, existing systems that combine these approaches share a critical blind spot: none condition their decisions on a *learned* representation of the current market regime. This gap defines the motivation for MARTA.

1.2 Host Organization

1.2.1 Presentation of Coral-io

Coral-io is a technology consulting firm specializing in digital transformation and IT solutions across multiple industries. The company provides comprehensive services that bridge technical expertise with business objectives, helping clients navigate complex technological challenges. Its activity spans five main areas:

- **DevOps Services**: CI/CD pipeline implementation, Infrastructure as Code, containerization and orchestration (Docker, Kubernetes), and test automation.
- **Application Performance Management (APM)**: full-stack observability, real-time monitoring, and bottleneck diagnosis for production software.

- **Third-Party Solution Integration**: as a Micro Focus (OpenText) partner, Coral-io delivers IT Operations Management (ITOM) integration and legacy system modernization.
- **Research & Development**: proof-of-concept development and emerging-technology adoption, sitting at the intersection of technical expertise and applied innovation.
- **Finance and Insurance Consulting**: FinTech implementation, regulatory compliance, and digital transformation for financial services clients.

MARTA was carried out within the R&D and Finance/Insurance Consulting scope of the company's activity, combining applied AI research with a financial-markets use case.

1.2.2 Internship Framework

Unlike a project built around a pre-defined internal need, this final-year project was conducted under a self-directed research framework: Coral-io granted full autonomy in selecting the subject and technical direction of the work, provided it aligned with the company's R&D and FinTech interests. This arrangement reflects a model already familiar from the author's prior internship experience at Coral-io, where independent, exploratory project work was similarly supported as a way to build internal expertise in emerging domains.

Within this framework, technical guidance was provided collaboratively by Ms. Oumaima Aounallah and the wider Coral-io team, consulted on an as-needed basis for architectural decisions, tooling questions, and periodic progress reviews rather than through prescriptive task assignment. Academic supervision was provided by Saoussen Lakhdhar at ESPRIT.

This structure placed a greater share of responsibility for scoping, planning, and technical decision-making on the author than is typical of a company-assigned project — a distinction elaborated further in Section 1.6.

1.3 Problem Statement

1.3.1 Limitations of Existing Approaches

Three broad families of automated trading systems exist today, each with a distinct blind spot relevant to MARTA.

Rule-based and classical statistical strategies (moving-average crossovers, mean-reversion bands, momentum filters) are transparent and computationally cheap, but their parameters are fixed at design time. When the market regime shifts — from trending to ranging, from low to high volatility — these fixed parameters no longer match the underlying data-generating process, and performance degrades until the strategy is manually re-tuned.

Pure reinforcement learning agents address this rigidity by learning a policy directly from historical price action, allowing some adaptation to changing dynamics. However, standard RL formulations treat the market as a single, undifferentiated environment: the agent is not explicitly told which regime it is currently operating in, and must infer this implicitly through its state representation. In practice, this means the

policy tends to average its behavior across regimes rather than specializing for each one, and offers no human-interpretable explanation for why a given action was chosen.

LLM-based trading agents, a more recent development, exploit large language models' ability to reason over unstructured text — news headlines, macroeconomic reports, analyst commentary — to produce qualitative market assessments. This adds a reasoning layer absent from purely numerical approaches, and can in principle justify its decisions in natural language. However, existing systems in this category typically condition the LLM's reasoning only on recent price statistics or raw text, without any structured, learned signal describing the broader statistical regime the market is currently in.

1.3.2 The Core Gap

Across all three families, no system combines (i) an explicit, learned representation of market regime, (ii) multi-perspective reasoning informed by that representation, and (iii) an execution layer that adapts its risk behavior according to regime and reasoning confidence. Regime is either ignored entirely, treated as a hidden variable an RL agent must discover implicitly, or left to ad hoc heuristics (e.g., a simple volatility threshold) rather than learned from data.

1.3.3 Formal Problem Statement

Given a multi-asset time series of market data, the problem addressed by this work is twofold:

1. **Regime representation learning**: learn a low-dimensional, temporally-aware embedding of market state that captures latent structure predictive of near-term dynamics, without relying on hand-crafted regime labels.

2. **Regime-conditioned decision making**: use this learned representation to condition both a multi-perspective reasoning process and a downstream execution policy, such that trading behavior adapts to the identified regime rather than applying a single fixed strategy across all market conditions.

Success is measured not only by trading performance (risk-adjusted returns, drawdown) but by whether regime conditioning demonstrably improves outcomes relative to regime-agnostic baselines — the central empirical question this report answers in Chapter **??**.

1.4 Proposed Solution

MARTA (Multi-Agent Regime-Aware Trading Architecture) addresses this gap through a five-layer pipeline, each layer directly targeting one part of the problem statement above.

1. **Regime Encoder**: a Temporal Fusion Transformer (TFT) trained with an InfoNCE contrastive objective learns dense embeddings of market state from hourly multi-asset data (BTC, ETH, EUR/USD, XAU/USD), without requiring hand-labeled regimes.

2. **Regime Detection**: a Gaussian Hidden Markov Model is fitted over the encoder's embedding space, discretizing the continuous representation into a small number of interpretable, statistically distinct regimes with associated transition dynamics.

3. **Multi-Agent Deliberation**: five specialized LLM agents debate the appropriate trading stance given the current regime, each contributing a different analytical perspective (e.g., trend, risk, contrarian), producing a reasoned, regime-aware recommendation rather than a black-box signal.

4. **RL Execution Supervisor**: a PPO-trained agent translates the agents' deliberation output into concrete position sizing and execution decisions, learning to modulate risk exposure according to regime and the confidence of the upstream reasoning.

5. **Monitoring Dashboard**: a Next.js interface exposes regime detection, agent reasoning, and execution decisions in real time, making the system's behavior auditable rather than opaque.

This layered design directly mirrors the two-part problem statement: layers 1–2 solve the regime representation learning problem, and layers 3–4 solve the regime-conditioned decision making problem. Layer 5 is a supporting component providing interpretability rather than a decision-making stage in its own right. The empirical contribution of regime-awareness across layers 1–4 is evaluated through dedicated ablation studies, presented in Chapter **??**.

1.5 Project Objectives and Scope

1.5.1 Objectives

The project pursues four concrete objectives, each corresponding to a layer of the proposed solution:

1. **Learn a meaningful regime representation**: train a TFT + InfoNCE encoder on multi-asset hourly data such that the resulting embedding space captures structure relevant to near-term market dynamics, validated through downstream regime separability rather than reconstruction loss alone.

2. **Discover interpretable, statistically distinct regimes**: fit a Gaussian HMM over the learned embeddings and verify that the resulting regimes are stable, distinguishable, and align with intuitive market conditions (e.g., trending vs. ranging, high vs. low volatility) upon inspection.

3. **Build a regime-aware multi-agent reasoning layer**: design and implement a five-agent LLM deliberation system whose recommendations are conditioned on detected regime, and evaluate whether this conditioning improves recommendation quality over a regime-agnostic baseline.

4. **Train an adaptive execution policy**: implement a PPO-based execution supervisor that translates agent deliberation into position-sizing decisions, and evaluate — via ablation — whether regime-awareness measurably improves risk-adjusted trading performance relative to an equivalent regime-agnostic system.

A secondary, supporting objective is to expose the full pipeline through a monitoring dashboard, so that regime detection, agent reasoning, and execution decisions remain auditable rather than opaque — a property largely absent from prior work in this space (Section 1.3.1).

1.5.2 Scope

In scope:

- Four assets: BTC/USD, ETH/USD, EUR/USD, XAU/USD, at hourly resolution, 2018–2025.
- Design, training, and evaluation of all five architectural layers described in Section 1.4.
- Backtesting and ablation studies quantifying the contribution of regime-awareness specifically.
- A functional monitoring dashboard for the full pipeline.

Out of scope:

- Live deployment with real capital; the system is evaluated via backtesting and paper-trading simulation only.
- Assets or timeframes beyond the four instruments and hourly resolution listed above.
- Regulatory, compliance, or brokerage-integration work required for production trading.
- Optimization of LLM inference cost/latency for high-frequency use cases; the multi-agent layer operates at hourly decision frequency, not tick-level.

This scope reflects the project's nature as an applied research contribution demonstrating the value of regime-awareness, rather than a production-ready trading product.

1.6 Project Methodology and Work Planning

Given the exploratory, research-oriented nature of MARTA, a rigid upfront specification was not appropriate: each architectural layer's design depended on empirical results from the layer built before it (e.g., the number and structure of latent regimes discovered by the HMM directly shaped how agent prompts were engineered). The project therefore followed an **iterative, layer-by-layer methodology**, where each phase produced a working, evaluable component before the next was started, rather than a big-bang integration at the end.

1.6.1 Development Phases

The work was organized into seven successive phases, summarized in Table 1.1 and Figure 1.1. The first phase, covering literature review, problem definition, and architecture design, produced Chapter 2's survey and the five-layer design introduced in Section 1.4; the remaining six phases correspond one-to-one with the architectural layers and validation work described in Chapter **??**.

Table 1.1: MARTA development phases

Phase	Deliverable	Duration & dates
1. Literature Review & Architecture Design	State-of-the-art survey, 5-layer design	12 wk (Mar 1 – May 24)
2. Data & Feature Engineering	Cleaned multi-asset hourly dataset (2018–2025)	2 wk (May 25 – Jun 7)
3. Regime Encoder	Trained TFT + InfoNCE encoder	3 wk (Jun 8 – Jun 28)
4. Regime Detection	Gaussian HMM over encoder embeddings	1 wk (Jun 29 – Jul 5)
5. Multi-Agent System	5-agent LLM debate pipeline	2 wk (Jul 6 – Jul 19)
6. RL Execution Agent	PPO supervisor trained on agent outputs	3 wk (Jul 20 – Aug 9)
7. Dashboard & Validation	Next.js dashboard, live system, ablations	3 wk (Aug 10 – Aug 31)

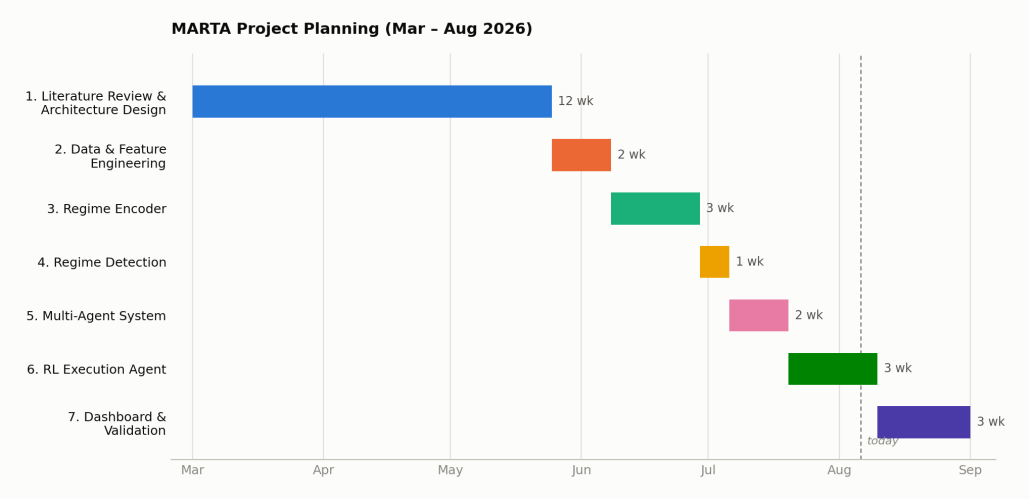


Figure 1.1: Project planning Gantt chart

1.6.2 Rationale for the Approach

This phase-by-phase structure offered two advantages over a monolithic design-then-build approach. First, it allowed early validation of the encoder and regime detection stages in isolation — verifying, for instance, that discovered regimes were interpretable and stable before committing to an agent architecture built on top of them. Second, it kept technical risk contained: had the HMM failed to produce meaningful regime clusters, the multi-agent and RL layers could have been redesigned without discarding the data pipeline or encoder work.

The sections of Chapter **??** follow this same phase ordering, each one corresponding to a development stage described above.

Chapter 2

State of the Art

2.1 Introduction

This chapter surveys the three bodies of work MARTA draws on and positions itself against: market regime detection, LLM-based trading agents, and reinforcement learning for trading. Section 2.2 briefly traces how algorithmic trading arrived at its current state. Sections 2.3–2.5 review each of the three families in turn, closing each with the specific limitation that motivates one part of MARTA's design. Section 2.6 then consolidates these gaps into a single comparative view and states precisely where MARTA sits relative to prior work.

2.2 Algorithmic Trading: Evolution and Challenges

Automated trading systems have progressed through three broad generations. The first generation consisted of rule-based strategies — moving-average crossovers, momentum filters, mean-reversion bands — valued for their transparency but structurally unable to adapt once market conditions diverge from the assumptions baked into their fixed parameters. The second generation applied supervised and reinforcement learning to trading directly, allowing strategies to be learned from data rather than hand-specified, at the cost of interpretability. The third and most recent generation incorporates large language models, exploiting their capacity to reason over unstructured text such as news and analyst commentary rather than numerical features alone. Each generation solved a limitation of the one before it while introducing a new one, a pattern examined in detail across the following three sections.

2.3 Market Regime Detection

2.3.1 Rule-Based Approaches

The simplest regime-detection methods rely on hand-crafted thresholds over indicators such as realized volatility or moving-average slope to classify the market into discrete states (e.g., trending vs. ranging). These approaches are computationally trivial and easy to interpret, but the thresholds themselves must be manually tuned per asset and require periodic re-calibration as market conditions evolve, making them brittle outside the conditions they were tuned on.

2.3.2 Statistical Approaches: Hidden Markov Models and Clustering

Hidden Markov Models (HMMs) have long served as the primary statistical tool for regime detection in finance, following the foundational regime-switching framework introduced by Hamilton for modeling business-cycle turning points in macroeconomic time series [**?**]. Applied to financial returns, an HMM treats the market regime as an unobserved latent state that generates observable return and volatility patterns, with transition probabilities between states estimated directly from data rather than imposed by rule. Gaussian HMMs fitted over hand-engineered feature sets (returns, realized volatility) have been shown to recover interpretable bull/bear/consolidation regimes on major equity indices, and have been used as risk-managing filters that suppress trading during predicted high-volatility states. K-means and Gaussian Mixture clustering offer a simpler, non-sequential alternative, but typically yield only two well-separated clusters (bullish/bearish) and lack the transition-probability structure that makes HMMs useful for characterizing regime persistence and switching dynamics.

The central limitation of this family, for MARTA's purposes, is the feature representation the HMM operates on: virtually all published work fits the HMM directly over hand-selected statistical features (returns, volatility, sometimes volume), rather than over a representation learned to capture broader temporal structure. This caps the ceiling of what the HMM can discover — it can only find regimes expressible in terms of the features it was given.

2.3.3 Deep Learning Approaches

More recent work combines regime detection with deep reinforcement learning rather than treating it as a standalone preprocessing step. A 2025 study integrating an HMM-based regime filter directly into the FinRL reinforcement learning framework found that regime-aware hybrid agents outperformed standard regime-agnostic FinRL baselines on risk-adjusted return metrics across a 30-stock portfolio, providing direct empirical support for the core hypothesis underlying MARTA: that explicit regime conditioning improves trading outcomes relative to regime-agnostic learning. However, this and comparable work still fits the HMM over hand-engineered statistical features rather than a learned temporal representation, and pairs regime detection only with an RL execution layer — it does not incorporate any reasoning layer capable of explaining *why* a regime implies a particular stance, which is precisely the gap MARTA's multi-agent layer is designed to fill.

2.4 LLM Agents in Finance

2.4.1 Single- and Multi-Agent LLM Trading Systems

TradingGPT, an early multi-agent system, assigns LLM agents layered memory and distinct behavioral characters to simulate diverse trader personas contributing to a shared decision. FinAgent, presented at KDD 2024, is a multimodal foundation agent that fuses textual news, numerical price data, and visual chart information through a dual-level reflection module, allowing the agent to adapt its strategy based on recent market feedback; evaluated across six financial datasets spanning equities and crypto, it sub-

stantially outperformed prior single-modality baselines. TradingAgents, proposed shortly after, more directly mirrors the organizational structure of a real trading firm: specialized analyst agents (fundamental, sentiment, technical) feed a research team of Bull and Bear debaters, whose output passes through a risk management team before a trader agent commits to a decision, and the resulting framework outperformed several classical baselines (Buy-and-Hold, MACD, RSI) on cumulative return and Sharpe ratio in simulated equity trading.

2.4.2 Multi-Agent Debate as a Design Pattern

The common thread across TradingGPT, FinAgent, and TradingAgents is the use of multiple specialized agents whose outputs are synthesized rather than a single monolithic LLM call — a design pattern motivated by the observation that decomposing analysis into distinct perspectives (technical, sentiment, risk) produces more robust recommendations than asking one model to reason about all of them simultaneously. MARTA's five-agent deliberation layer adopts this same debate pattern.

2.4.3 Limitations of Existing Approaches

None of the systems reviewed above condition their agents' reasoning on a learned, structured representation of market regime. TradingAgents' agents reason over recent price statistics and text directly; FinAgent's reflection module adapts based on recent performance feedback rather than an explicit regime signal; TradingGPT's characters are fixed personas rather than regime-conditioned roles. In each case, if the notion of "market regime" appears at all, it is implicit in whatever the LLM infers from raw inputs at inference time, not an explicit, pretrained signal shared across agents. This is the specific gap MARTA's regime encoder and HMM layer are designed to fill upstream of the agent debate.

2.5 Reinforcement Learning for Trading

2.5.1 Early Approaches: DQN and Discrete Action Spaces

Early applications of deep RL to trading adapted Deep Q-Networks (DQN) to discrete action spaces (buy/sell/hold), following the broader deep RL literature. FinRL, an open-source framework from Columbia University and NYU Shanghai, standardized this line of work by providing a full pipeline — market environments, fine-tuned implementations of DQN, DDPG, PPO, SAC, A2C, and TD3, and common reward functions — specifically to lower the engineering barrier for applying deep RL to quantitative trading. DQN-based agents are conceptually simple but constrained to discrete position sizes, limiting their ability to express graduated risk exposure.

2.5.2 PPO and Continuous Action Spaces

Proximal Policy Optimization (PPO), introduced by Schulman et al., addresses the instability of earlier policy gradient methods through a clipped surrogate objective that constrains the size of each policy update, achieving much of the stability of Trust Region Policy Optimization while being substantially simpler to implement. Its compatibility

with continuous action spaces makes it well suited to position-sizing tasks, where the agent must output a continuous fraction of capital to allocate rather than a discrete action, and it is the algorithm implemented for MARTA's execution supervisor for exactly this reason.

2.5.3 Curriculum Learning in Finance

Curriculum learning — structuring training so an agent first masters a simpler version of a task before progressing to a harder one — is comparatively underused in financial RL relative to other RL application domains, though the general principle (start with a single, liquid, well-behaved asset before extending to a more volatile or lower-liquidity one) aligns with standard practice in curriculum design more broadly. MARTA's own BTC-to-BTC+ETH curriculum training, detailed in Chapter **??**, follows this logic.

2.6 Positioning of MARTA

Table 2.1 consolidates the gaps identified across the three families reviewed above.

Table 2.1: Comparison of MARTA against representative prior systems

System	**Learned regime repr.**	**Multi-agent reasoning**	**Adaptive executi**
HMM regime filters [**?**]	Partial (hand-features)	No	No
FinRL [**?**]	No	No	Yes (RL)
HMM + FinRL [**?**]	Partial (hand-features)	No	Yes (RL)
TradingGPT	No	Yes	No
FinAgent [**?**]	No	Partial (single agent)	No
TradingAgents [**?**]	No	Yes	No
MARTA (this work)	**Yes (TFT + InfoNCE)**	**Yes (5 agents)**	**Yes (PPO)**

No system reviewed combines a learned regime representation, multi-agent reasoning conditioned on that representation, and an adaptive execution layer in a single pipeline. This is precisely the combination MARTA proposes: a TFT encoder trained with an InfoNCE contrastive objective [**?**, **?**] learns the regime representation that HMM-based approaches instead derive from hand-selected features; this representation conditions the multi-agent debate that LLM trading systems otherwise perform without any explicit regime signal; and a PPO-based execution supervisor [**?**] translates the resulting recommendation into position sizing, closing the loop that regime-detection-only systems leave open.

2.7 Summary

This chapter reviewed market regime detection, LLM-based trading agents, and reinforcement learning for trading as three largely separate research threads, each addressing part of the problem MARTA targets while leaving a specific gap: regime detection methods rely on hand-crafted features rather than learned representations; LLM agent systems reason without any explicit regime conditioning; and RL execution methods, even when regime-aware, lack an interpretable reasoning layer. Chapter **??** now turns to how MARTA's

five-layer architecture, introduced in Section 1.4, is designed and implemented to close these gaps jointly.



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